

College Student Dropout Prediction.

1. Problem Statement:

- Students who leave college before completing a degree may lose time, money, and access to opportunities.
Academic advisers must decide how to direct limited support toward students who may need it. This project will examine whether student records can help identify those at risk of leaving their program. Students, advisers, and retention staff could benefit from earlier invitations to tutoring, financial guidance, or academic planning. Predictions would support a conversation with a student, rather than determine admission, financial aid, or access to services

2. Objective:

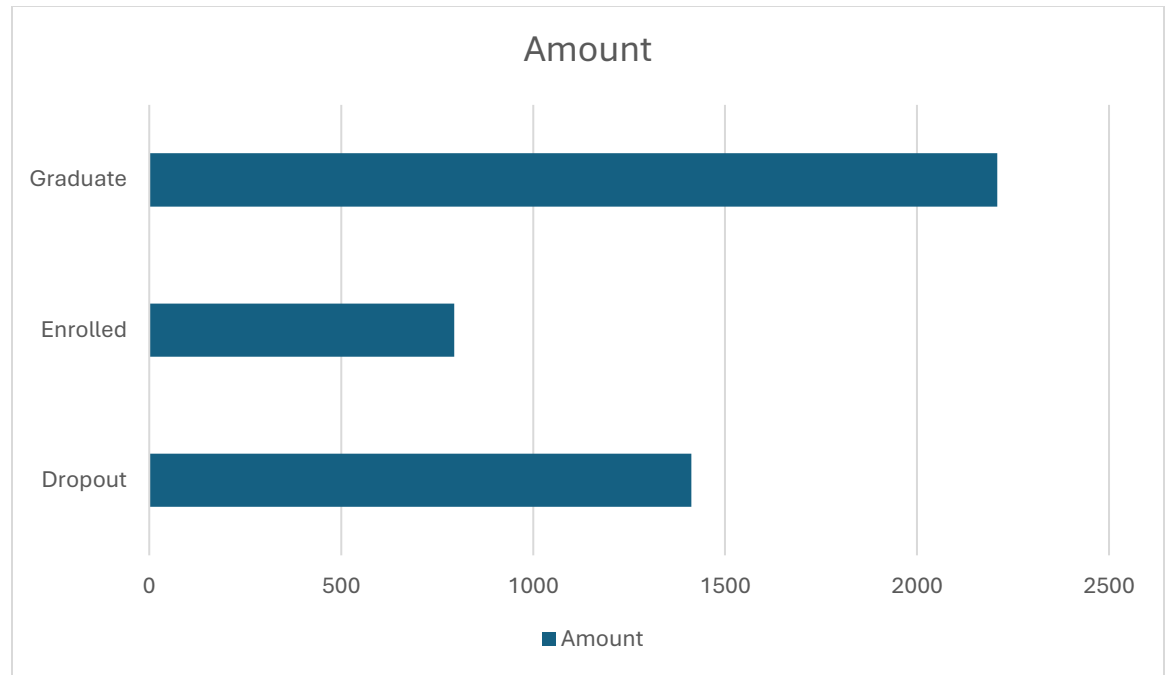
- We will predict whether a student will be classified as Dropout, Enrolled, or Graduate at the end of the normal program duration using admission, prior education, and demographic information available at enrollment

3. Dataset:

- Source URL: UCI, Predict Students' Dropout and Academic Success: <https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>
- License: Creative Commons Attribution 4.0 International (CC BY 4.0); attribution is required.
- Row count / column count: 4,424 student records; 37 columns, comprising 36 predictors and one target, verified from the downloaded UCI file.
- One row: One undergraduate student at the Polytechnic Institute of Portalegre, Portugal.
- Collection: The authors combined institutional academic records and teaching-support data, national admission records from DGES/CNAES, and macroeconomic data from PORDATA. Records cover entry years 2008/2009 through 2018/2019 across 17 programs. They merged and cleaned the records, addressed anomalies and missing values, and assigned outcomes at the normal program endpoint.

4. Target Variable:

- Column: Target. Type: nominal categorical text with three classes. Enrolled denotes continued enrollment at the endpoint; it is not a confirmed eventual dropout.
- Dropout: 1412, 32%
- Enrolled: 794, 18%
- Graduate: 2209, 50%



5. Feature Inventory:

Column name	Data type	Percent missing	Description
Marital status	Integer/category	0.00%	Marital-status category.
Application mode	Integer/category	0.00%	Admission route.
Application order	Integer/ordinal	0.00%	Application preference rank (0-9).
Course	Integer/category	0.00%	Degree-program code.
Daytime/evening attendance	Integer/binary	0.00%	Attendance: daytime 1, evening 0.
Previous qualification	Integer/category	0.00%	Prior education category.
Previous qualification (grade)	Real	0.00%	Prior qualification score (0-200).
Nationality	Integer/category	0.00%	Nationality code.
Mother's qualification	Integer/category	0.00%	Mother's education category.
Father's qualification	Integer/category	0.00%	Father's education category.
Mother's occupation	Integer/category	0.00%	Mother's occupation code.
Father's occupation	Integer/category	0.00%	Father's occupation code.

Admission grade	Real	0.00%	Admission score (0-200).
Displaced	Integer/binary	0.00%	Recorded displaced status (1 = yes).
Educational special needs	Integer/binary	0.00%	Special educational needs (1 = yes).
Debtor	Integer/binary	0.00%	Recorded debtor status (1 = yes).
Tuition fees up to date	Integer/binary	0.00%	Tuition payments current (1 = yes).
Gender	Integer/binary	0.00%	Recorded gender: male 1, female 0.
Scholarship holder	Integer/binary	0.00%	Scholarship recipient (1 = yes).
Age at enrollment	Integer	0.00%	Age in years at enrollment.
International	Integer/binary	0.00%	International student (1 = yes).
Curricular units 1st sem (credited)	Integer	0.00%	1st-semester units credited.
Curricular units 1st sem (enrolled)	Integer	0.00%	1st-semester units enrolled.
Curricular units 1st sem (evaluations)	Integer	0.00%	1st-semester evaluation count.
Curricular units 1st sem (approved)	Integer	0.00%	1st-semester units passed.
Curricular units 1st sem (grade)	Real	0.00%	1st-semester mean grade (0-20).
Curricular units 1st sem (without evaluations)	Integer	0.00%	1st-semester units not evaluated.
Curricular units 2nd sem (credited)	Integer	0.00%	2nd-semester units credited.
Curricular units 2nd sem (enrolled)	Integer	0.00%	2nd-semester units enrolled.
Curricular units 2nd sem (evaluations)	Integer	0.00%	2nd-semester evaluation count.
Curricular units 2nd sem (approved)	Integer	0.00%	2nd-semester units passed.
Curricular units 2nd sem (grade)	Real	0.00%	2nd-semester mean grade (0-20).
Curricular units 2nd sem (without evaluations)	Integer	0.00%	2nd-semester units not evaluated.
Unemployment rate	Real	0.00%	Unemployment rate (%).
Inflation rate	Real	0.00%	Inflation rate (%).
GDP	Real	0.00%	GDP indicator; UCI does not state units.
Target	Text/category	0.00%	Outcome: Dropout, Enrolled, Graduate.

6. What success looks like:

- Our model will be useful if it helps academic advisers identify students who may drop out early enough to offer support, such as tutoring, financial guidance, or academic planning. We will assess whether its predictions match students' recorded outcomes on data it has not seen before and whether it identifies students who need attention without overwhelming advisers with unnecessary alerts. A useful prediction would prompt a supportive conversation, not label a student as certain to fail. Missing a student who later drops out would generally be more harmful than incorrectly flagging a student who stays enrolled, because the missed student could lose an opportunity to receive help. However, unnecessary alerts could waste advisers' time or worry students, so predictions should guide careful, respectful outreach.

7. Risks:

- **Using information unavailable at enrollment:** Semester grades and course completion records could make predictions appear more accurate even though advisers would not have that information when students first enroll. We will exclude semester results from the enrollment model and use other variables only when their availability at enrollment can be verified.
- **Limited generalizability and potential bias:** The dataset represents historical students at one institution in Portugal, so patterns may not apply to students at other colleges or today. Demographic information could also produce uneven prediction quality across student groups. We will examine these differences where sample sizes allow and treat predictions as guidance for supportive outreach rather than grounds for restricting opportunities.